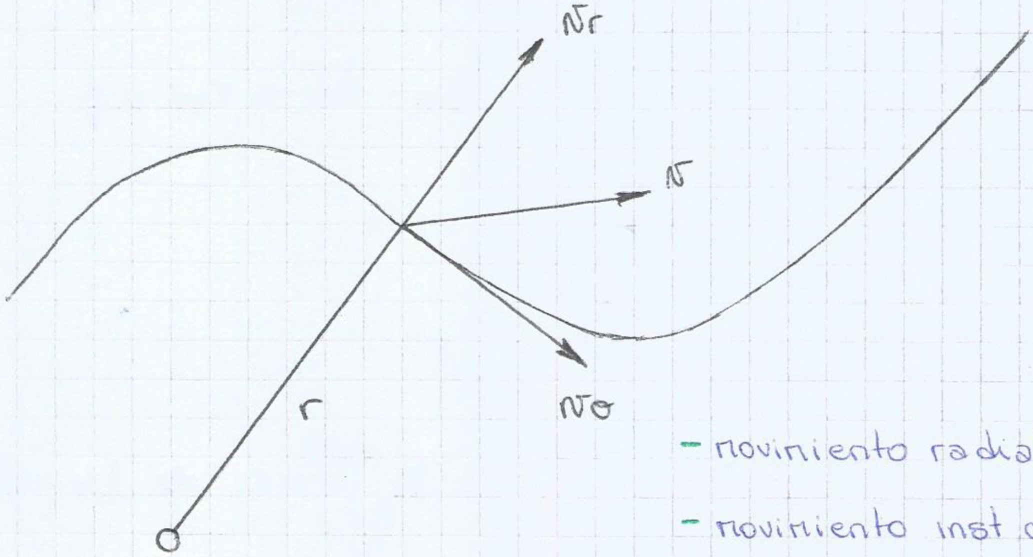
## Unidad VI: Impulso angular:




- movimiento radial $v_r$
- movimiento inst circular $v_\theta$

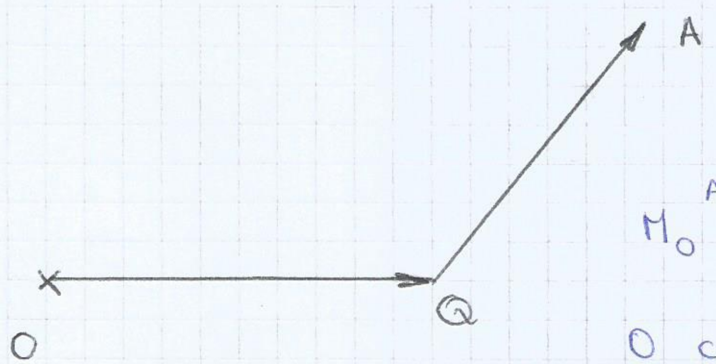
$$v = v_r + v_\theta$$

$$P = m v_r + m v_\theta$$

$$L_O = r \times p = r \times (m v_r + m v_\theta)$$
$$= r \times m v_r + r \times m v_\theta$$
$$= r m v_\theta$$

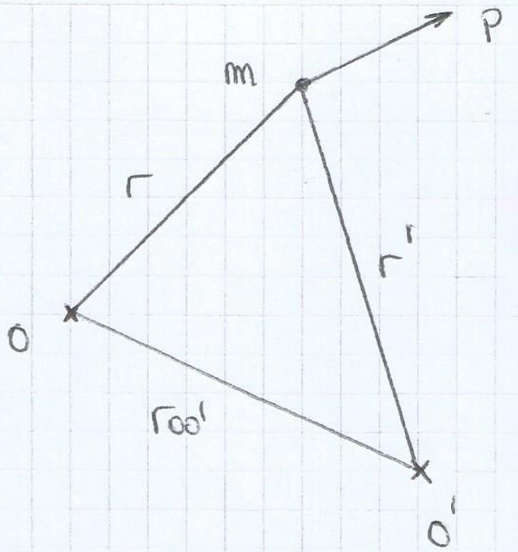
$L_O = r \times p$ impulso angular respecto de O.

## Momento de un vector A respecto de O.




$$M_O^A = r_{OQ} \times A$$

O centro de momentos



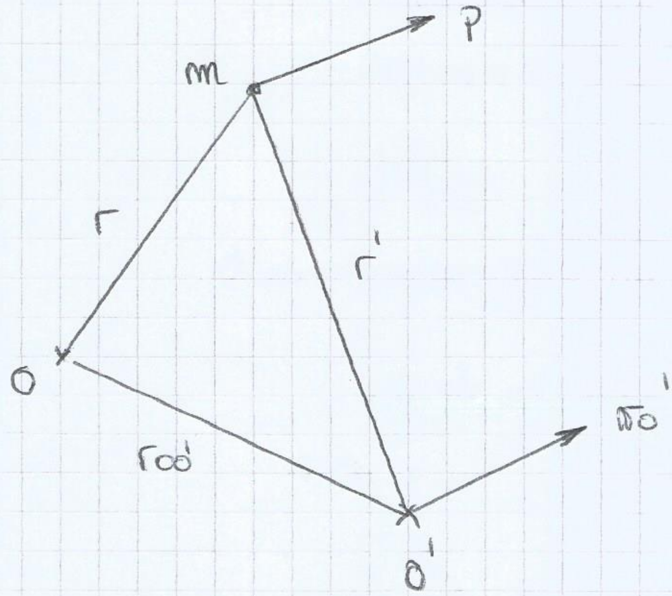

$$L_O = r \times p$$

$$L_O = (r' + r_{OO'}) \times p$$

$$L_O = L_{O'} + r_{OO'} \times p$$

$$\left.\begin{aligned} L_O &= r \times p \\ \frac{dL_O}{dt} &= \frac{dr}{dt} \times p + r \times \frac{dp}{dt} \\ \frac{dL_O}{dt} &= r \times F^{ext} = M_O^{ext} \end{aligned}\right\} \text{Si } M_O^{ext} = 0 \Rightarrow L_O = \text{cte.}$$

Supongamos que O' se mueve.




$$L_{O'} = L_O - r_{OO'} \times p$$

$$\frac{dL_{O'}}{dt} = \frac{dL_O}{dt} - \frac{dr_{OO'}}{dt} \times p - r_{OO'} \times \frac{dp}{dt}$$

$$\frac{dL_o}{dt} = M_o^{ext} - v_{o'} \times p - r_{oo'} \times F^{ext}$$

$$\frac{dL_o}{dt} = r \times F^{ext} - r_{oo'} \times F^{ext} - v_{o'} \times p$$

$$\frac{dL_o}{dt} = r' \times F^{ext} - v_{o'} \times p$$

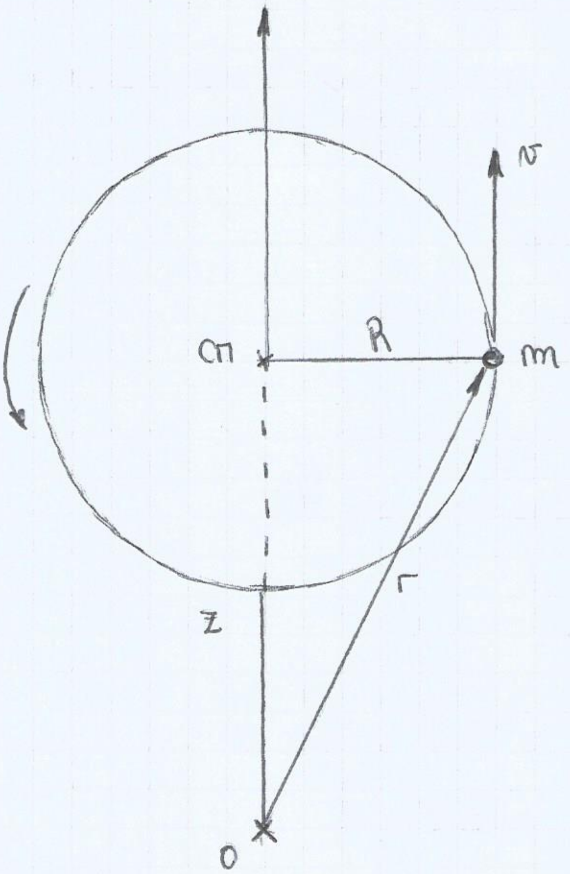
$$\frac{dL_o}{dt} = M_{o'}^{ext} - v_{o'} \times p$$

$v_o \times p = 0$ si $v_{o'} = 0$ Si $O' \equiv CM$

$v_{CM} \times p = v_{CM} \times M \, v_{CM} = 0$

Ley de conservación de $L_o$:

Si $M_o^{ext} = 0 \Rightarrow L_o = cte$ $\begin{cases} \text{Si } O \text{ está quieto} \\ \text{Si } O \equiv CM. \end{cases}$




$L_o = r \times p$ y $p = m v$

$v = \omega \times r = \omega \hat{z} \times R \hat{r}$

$v = \omega \times r = \omega R \hat{\theta}$

$r = z + r = z\hat{z} + R\hat{r}$ $p = m\omega R\hat{\theta}$

$L_o = (z\hat{z} + R\hat{r}) \times (m\omega R\hat{\theta})$

$L_o = -m\omega R z \hat{r} + m\omega R^2 \hat{z}$

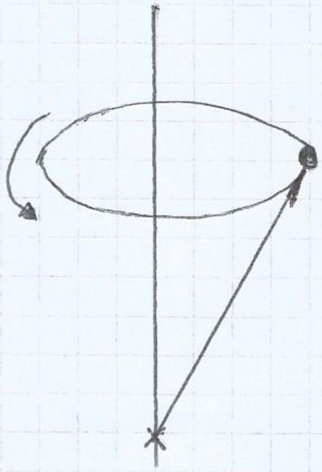
$L_o = L_{oz} + L_{or} = cte$

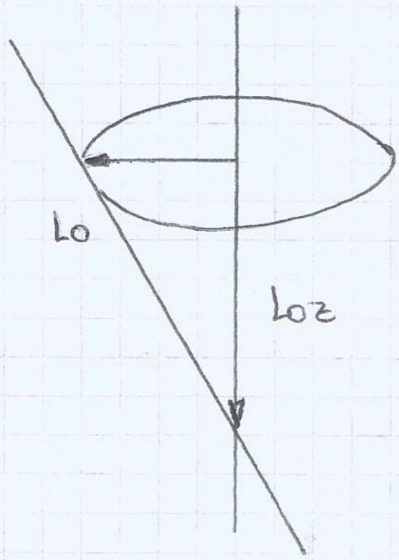
Si $\omega = cte$ $\quad L_{oz} = m\omega R^2$

$$|L_o| = \sqrt{m^2\omega^2R^2Z^2 + m^2\omega^2R^4}$$

$$|L_o| = \left(m^2\omega^2 R \left(\frac{R^2+Z^2}{r^2}\right)^{1/2}\right)$$

$$|L_o| = m\omega R = cte$$






$O \equiv CR$

Si $O \equiv CR \Rightarrow Z = 0$ y $L_{OR} = 0$

$L_{CR} = m\omega R^2 \hat{Z} = cte$ si $\omega = cte$

$$\frac{dL_{CR}}{dt} = m\dot{\omega} R^2 \hat{Z} = m\gamma R^2 \hat{Z} = M_{CR}^{ext}$$

$$F^{ext} = \underbrace{F_r \hat{r}}_{\text{Movimiento Circular}} + F_\theta \hat{\theta} \quad \text{(No hay en } \hat{z}\text{)}$$

$$M_{CR}^{ext} = R\hat{r} \times (F_r\hat{r} + F_\theta\hat{\theta}) = R\hat{r} \times F_\theta\hat{\theta}$$

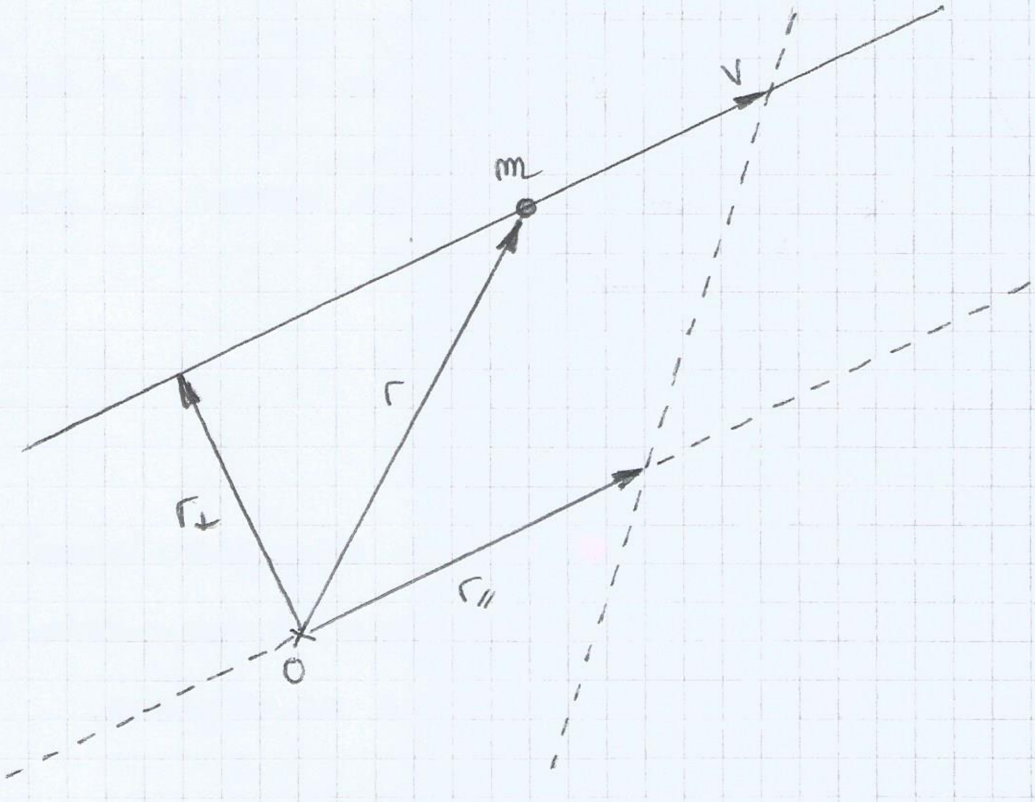
$$= R F_\theta \hat{Z}$$

Si $\omega = cte \Rightarrow M_{CR} = 0$

Si $O \neq CR$ , $\omega = cte \Rightarrow \frac{dL_O}{dt} = -m \omega R z \dot{\hat{r}}$

$$\frac{dL_O}{dt} = -m \omega R z \dot{\theta} \hat{\theta}$$

$$\frac{dL_O}{dt} = -m \omega^2 R z \hat{\theta}$$




$L_O = r \times m v$

$L_O = (r_{//} + r_\perp) \times m v$

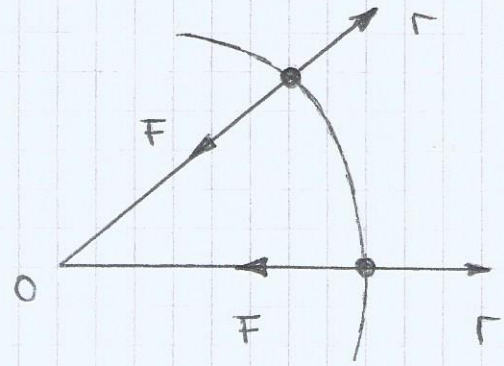
$L_O = m r_\perp v = cte$

Si $L_O = 0$ , caracteriza el movimiento de traslación.

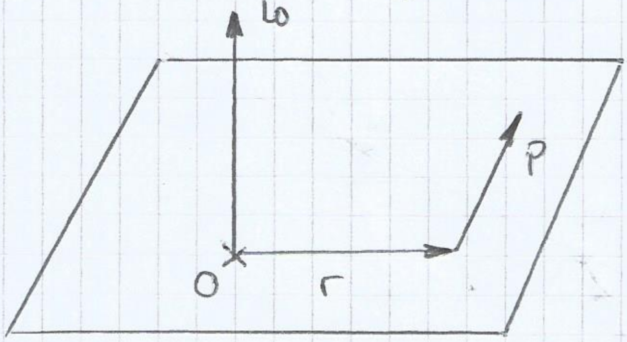
$L_O = r \times p$

$M_O^{ext} = 0 \iff L_O = cte$ sii $\begin{cases} O \text{ está quieto} \\ O \equiv CM \end{cases}$

$M_O^{ext} = r \times \bar{F}^{ext} = 0$




- $r = 0$
- $\bar{F}^{ext} = 0$
- $r \parallel \bar{F}^{ext}$ (Fuerzas centrales)




$L_O = r \times p = L_O \hat{n}$

$\hat{n}$ versor $\perp$ plano

Si se conserva:

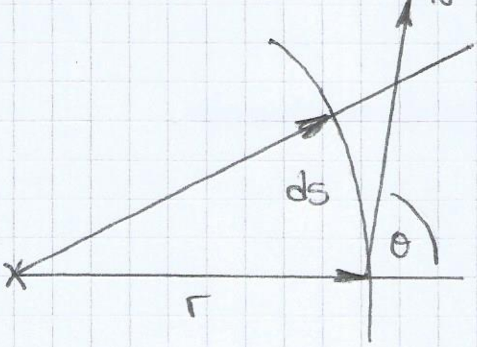
$$L_O = L_O'$$

- $|L_O| = |L_O'|$
- $\hat{n} = \hat{n}'$

$L_O' = r' \times p' = L_O' \hat{n}'$

- Cuando el impulso lineal se conserva, el movimiento se realiza en un plano.

## Ley de las áreas (2da ley de Kepler)



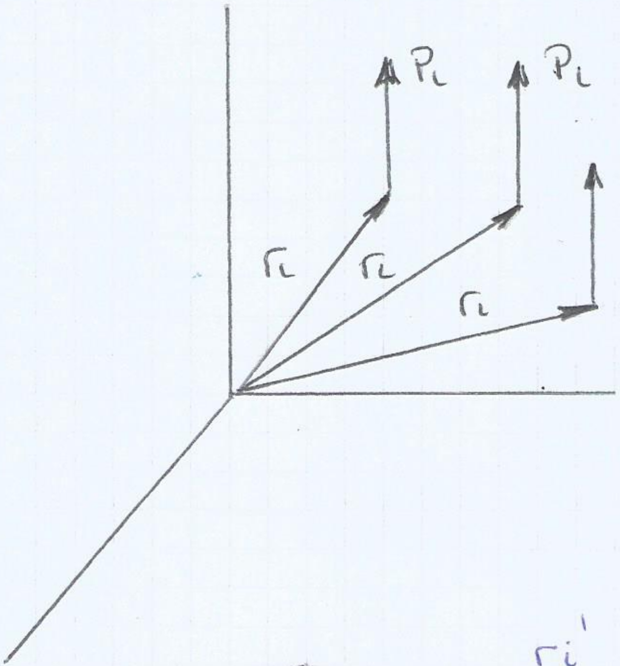

$|L_O| = |m r \times v| = m r v \operatorname{sen}\theta$

$dA = \frac{1}{2} r \, ds \operatorname{sen}\theta$

$dA = \frac{1}{2} r \operatorname{sen}\theta \, v \, dt$

$\frac{dA}{dt} = \frac{1}{2} r v \operatorname{sen}\theta = \frac{1}{2} \frac{|L_O|}{m} = cte$

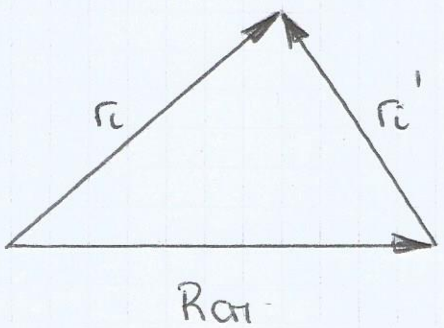
## Sistema de N partículas:




$$L_o = \sum_{i=1}^{N} r_i \times p_i$$

$$\sum_{i=1}^{N} (r_i - R_{cm}) \times p_i + \dots$$

$$\dots \sum_{i=1}^{N} R_{cm} \times p_i$$




$r_i'$ posición de la partícula $i$ respecto del centro de masa

$$L_o = \underbrace{L_{cm}} + \underbrace{R_{cm} \times p}$$

Momento angular intrínseco o de spin.

Como se mueven las partículas alrededor de CM

Como se mueve el CM alrededor del centro de momentos